Assignment 2

Group 8

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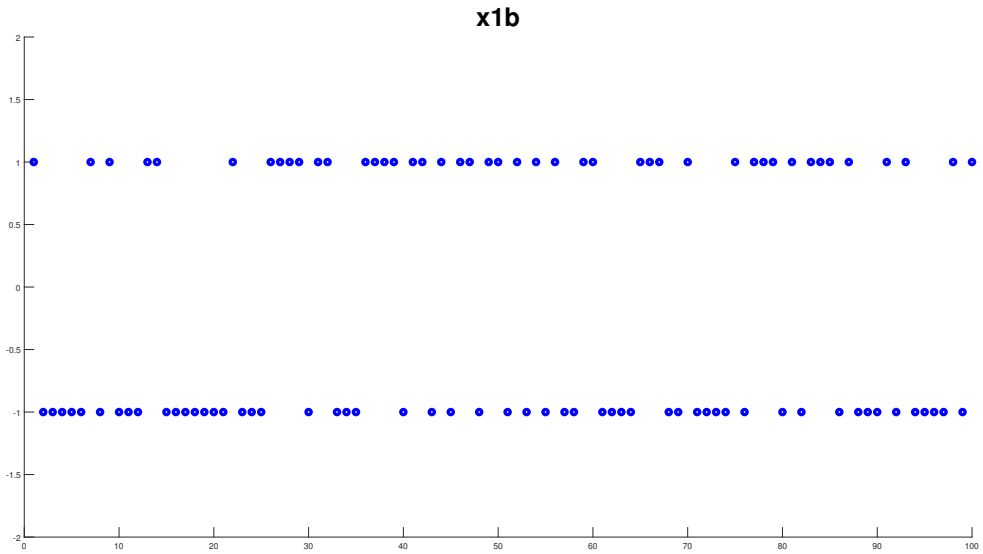
Roadmap

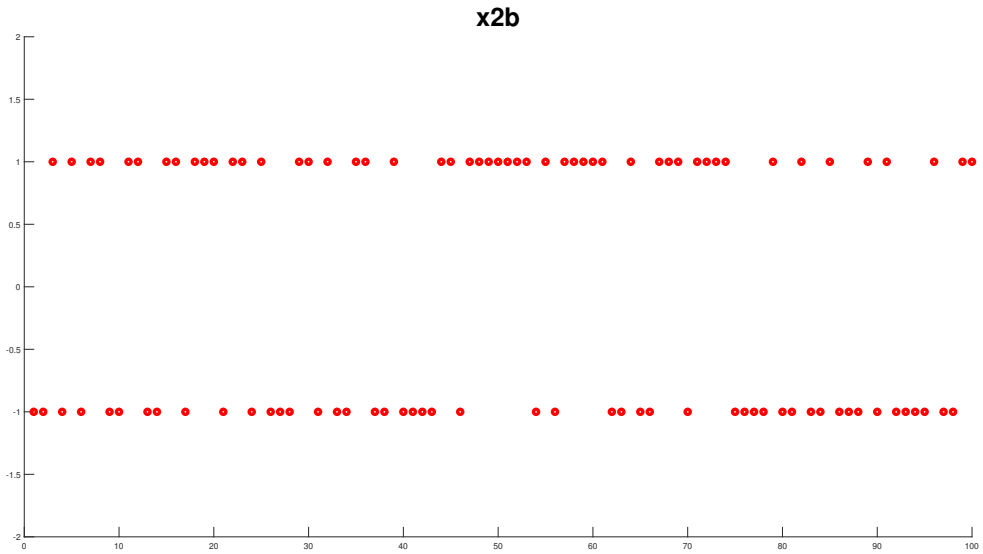
- Task 1: Gold code properties
- Task 2: Estimation of a single satellite delay
- Task 3: Performance with observed satellites
- Task 4: Performance under jamming attacks
- Task 5: Doppler impact

Task 1: Sequence Generation

Plot the first 100 bipolar values of the two sequences

Task 1: Sequence Generation





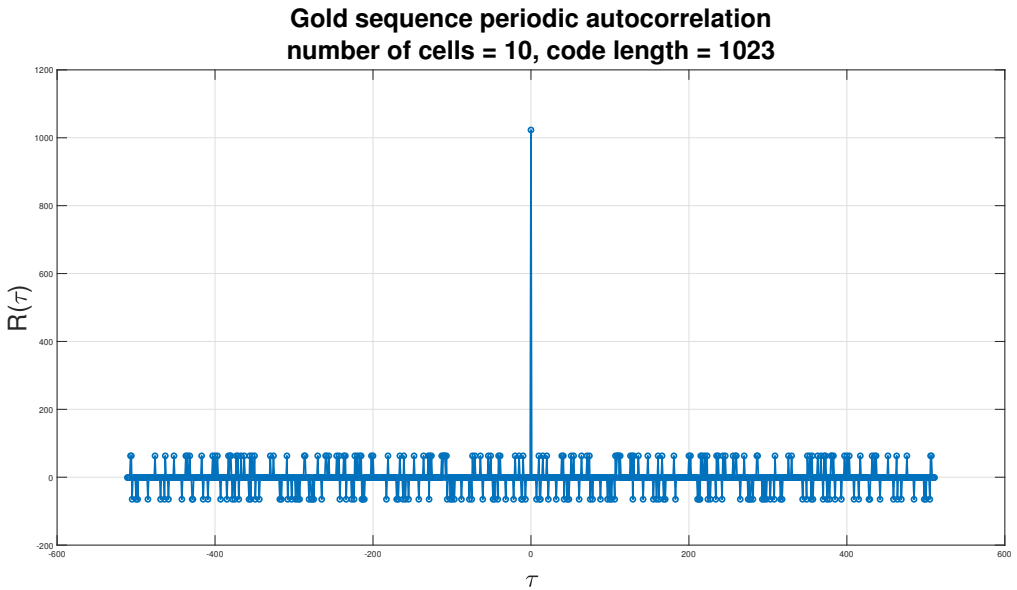
Task 1: Sequence Generation

The plots display the first 100 bipolar values for sequences $x1b$ and $x2b$, generated using specific LFSR polynomials and initial seeds. These sequences fluctuate between $+1$ and -1, representing the physical signal mapped from the underlying binary Gold codes. This visualization confirms the pseudo-random nature of the signals used to identify different satellites.

Task 1: Auto-correlation

Compute the periodic auto-correlation AC11 of x1b. Plot it centered at zero.

Task 1: Auto-correlation

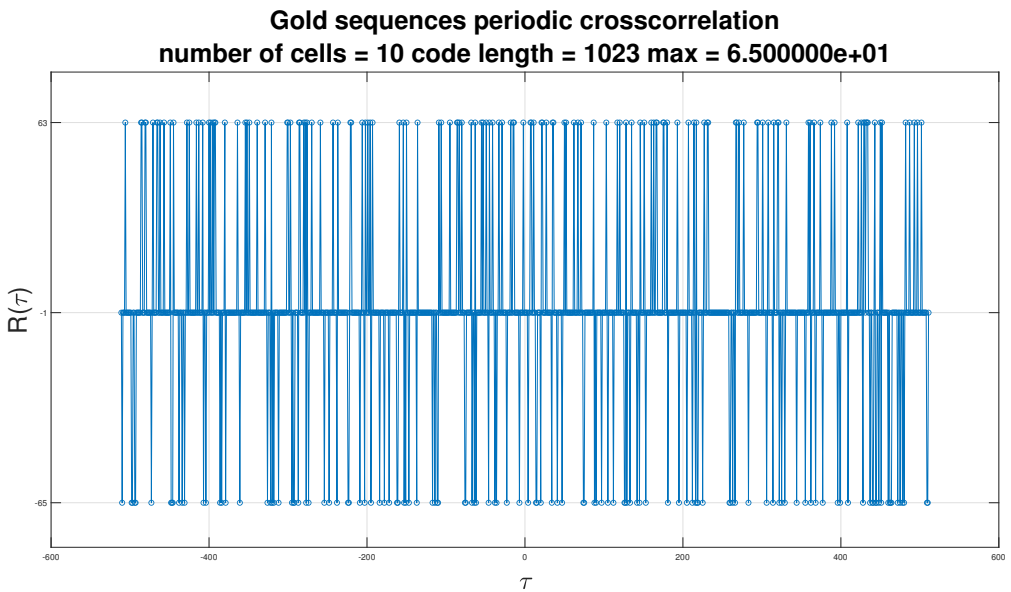


The auto-correlation plot for the $L = 1023$ sequence displays a sharp central peak of 1023 at zero lag, with exceptionally low side lobes. This near-orthogonality to its shifted replicas is crucial for a receiver to accurately estimate signal delays.

Task 1: Cross-correlation

Compute the periodic cross-correlation CC12 between x1b and x2b. Write the value of the maximum of the absolute value of the cross-correlation.

Task 1: Cross-correlation



The cross-correlation plot for the $L = 1023$ sequences shows a maximum absolute value of 65. This confirms the Gold codes' three-value property and guarantees low interference, allowing the receiver to reliably distinguish between different satellites.

Task 1: Three-value Property

Verify that the cross-correlation satisfies the three-value property of the Gold sequences.

Task 1: Three-value Property

We first calculated the theoretical three values of the gold sequence with m=10 by using the formula $t = 2^{((m+2)/2)} + 1$ to get three values: $[-t, -1, t-2]$ obtaining $[-65, -1, 63]$. We then proceeded to extract the unique values of the cross-correlation obtaining the same three values, verifying the three-value property of the Gold sequences.

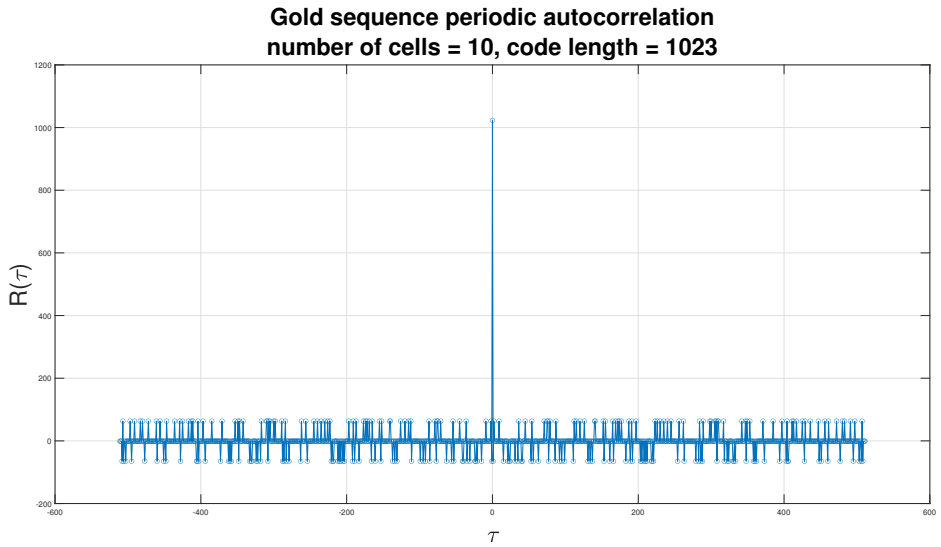
```
--- Verifying the three values property ---
Theoretical values:
   -65    -1    63

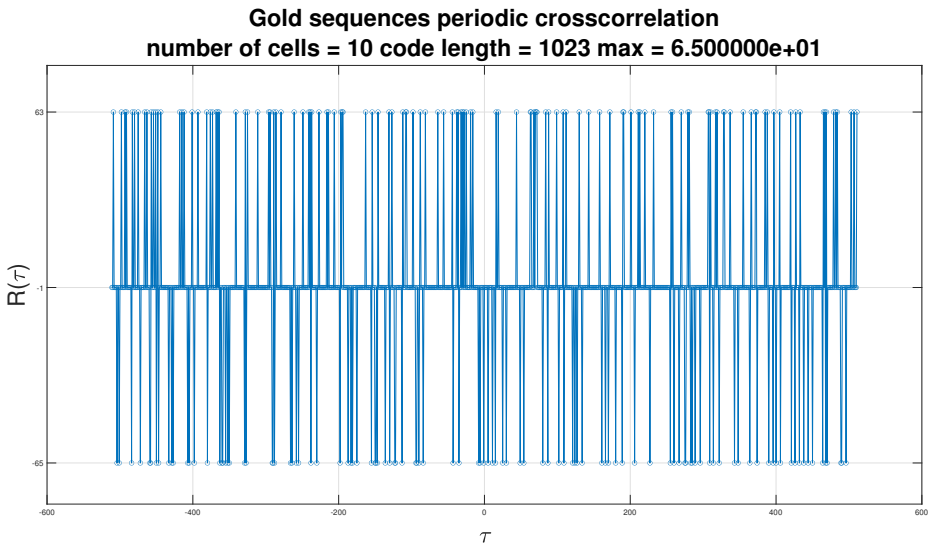
Unique sperimental values (CC12):
   -65    -1    63
```

Task 1: Seed Modification

Change the values of the starting seeds (FirstInitialConditions, SecondInitialConditions), write down their values, plot the auto-correlation and cross-correlation functions, and verify if all the properties are preserved.

Task 1: Seed Modification





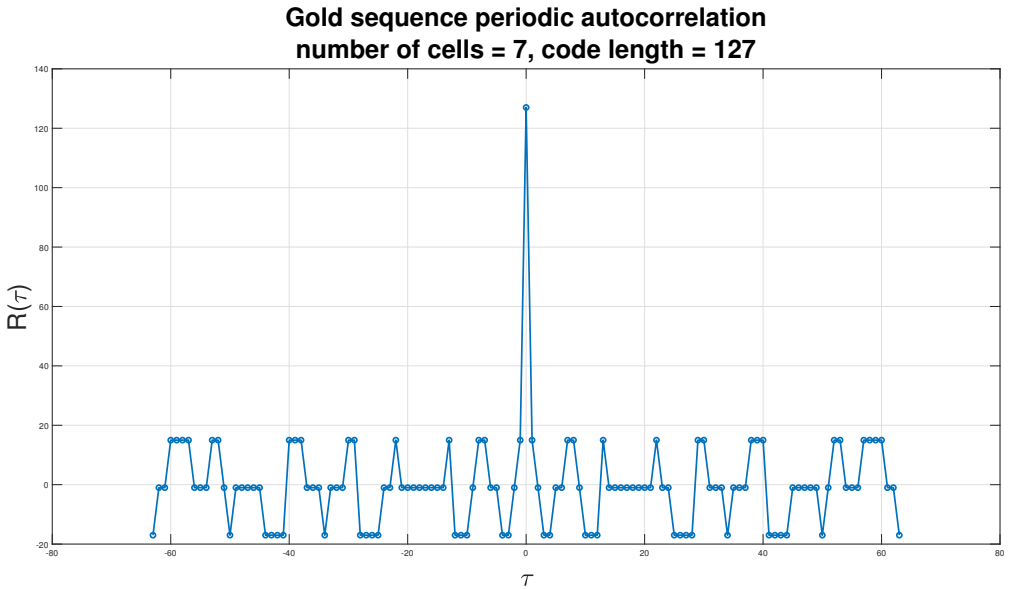
Task 1: Seed Modification

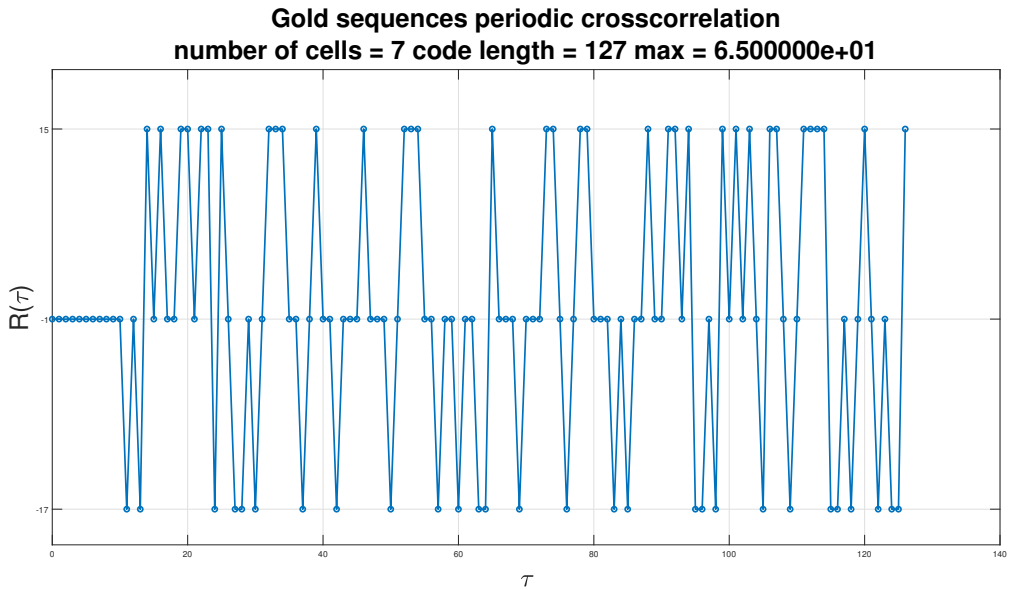
We changed the two seeds values to [1101001011] and [0110110010] and as we can see from the images the autocorrelation still presents a single peak at zero lag with a value equal at L (1023) and also that the three-value property is fully preserved.

Task 1: Length 127 Analysis

Repeat for two Gold codes with length 127.

Task 1: Length 127 Analysis





Task 1: Length 127 Analysis

The simulation for $L = 127$ confirms the three-value property with experimental values $\{-17, -1, 15\}$ matching theory. While the auto-correlation peak reaches 127, the lower peak-to-interference ratio of 7.4 indicates reduced isolation compared to longer codes. This makes the system more vulnerable to Multiple Access Interference, as the cross-correlation floor can more easily mask weaker signals.

```
--- Verifying the three values property (Cross-correlation L=127) ---
Theoretical Values:
   -17    -1    15

Unique sperimental values (CC12_127):
   -17    -1    15
```

Task 1: Comparison and Consequences

Compare the auto-correlation peak for $\tau = 0$ and the maximum of the cross-correlation values for $\tau \neq 0$ for two cases of length 1023 and 127. Making reference to how GPS works (slide 12), discuss the possible consequences.

Task 1: Comparison and Consequences

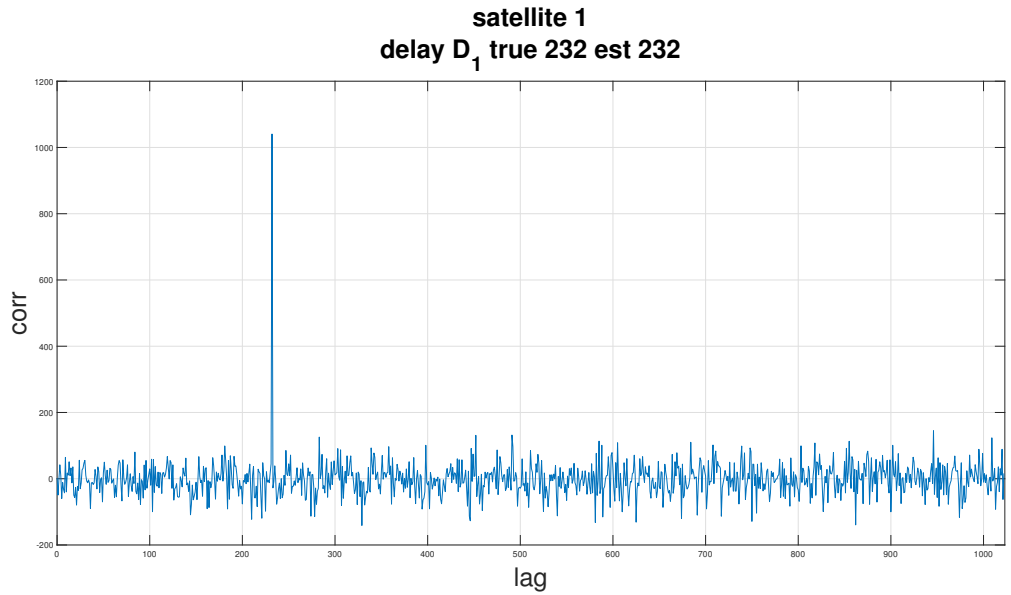
Comparing $L = 1023$ vs. $L = 127$:
Comparing the two sequence lengths reveals a significant difference in isolation capabilities. For $L = 1023$, the ratio between the auto-correlation peak (1023) and the maximum cross-correlation (65) is roughly 15.7. For $L = 127$, the auto-correlation peak is 127 and the maximum cross-correlation is 17, yielding a ratio of about 7.4.

Consequences for GPS:
The receiver distinguishes different satellites by exploiting these low cross-correlation values. A lower ratio implies worse interference rejection. If the cross-correlation noise floor is too high, a strong signal from a nearby satellite could easily mask the weaker signal of a distant one, severely degrading the receiver's positioning accuracy.

Task 2: Delay Estimation

- Compute the correlation between r and x1b and estimate the delay D1 (in samples).
- Plot the output of your correlation, and write the true and estimated values of D1 above the figure. Verify that they match.

Task 2: Delay Estimation Results



As illustrated in the figure, there is a perfect match between the true delay and the estimated peak. The correlation reliably detects the correct delay, which in this trial is successfully estimated at 232 samples.

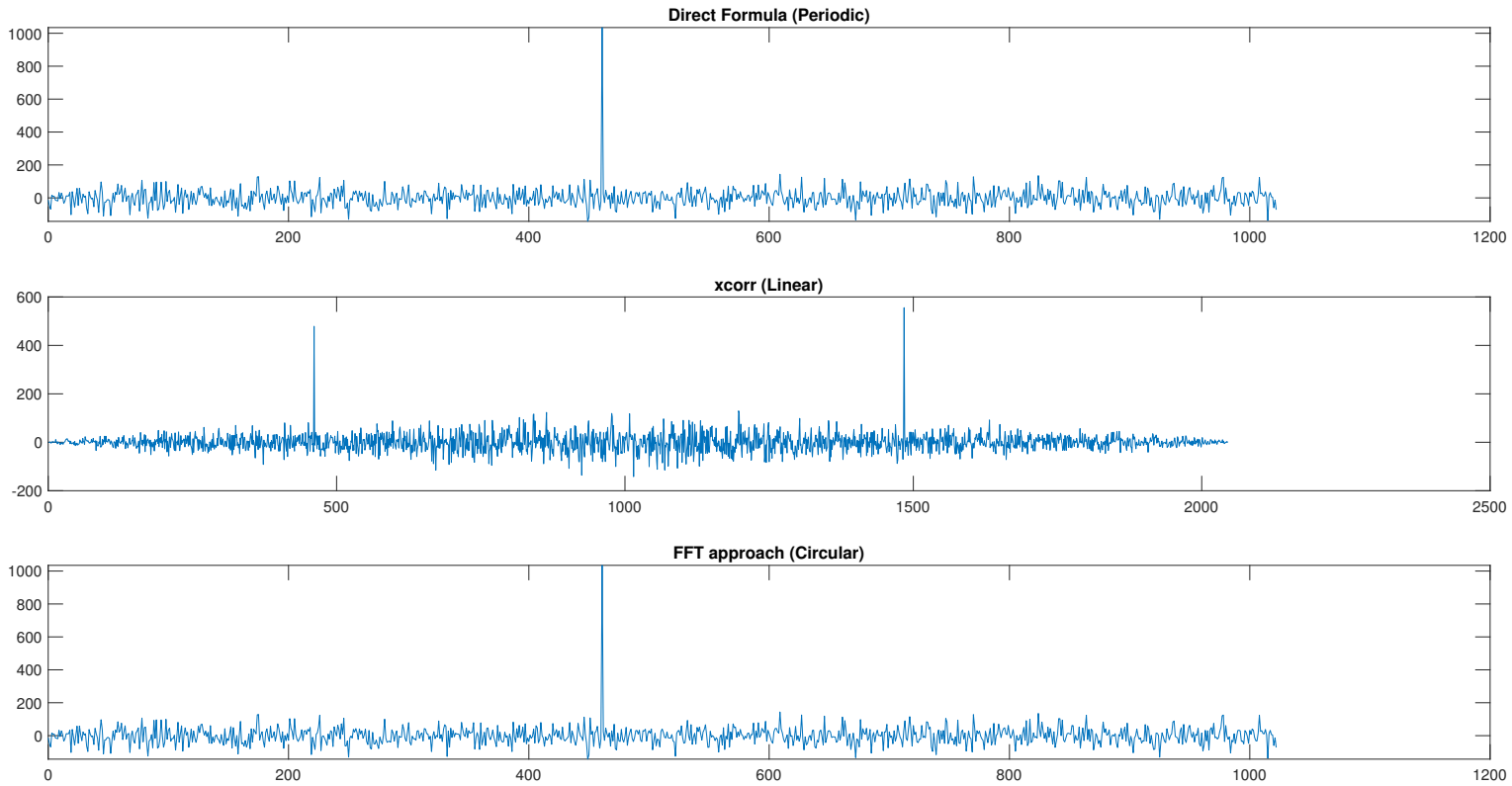
Task 2: Correlation Techniques

Compare different techniques for computing the correlation including at least

- direct formula using ($r. * xx1b$)
- xcorr
- fft

Plot the different correlation profiles, and discuss them (important: point out the differences)

Task 2: Correlation Techniques



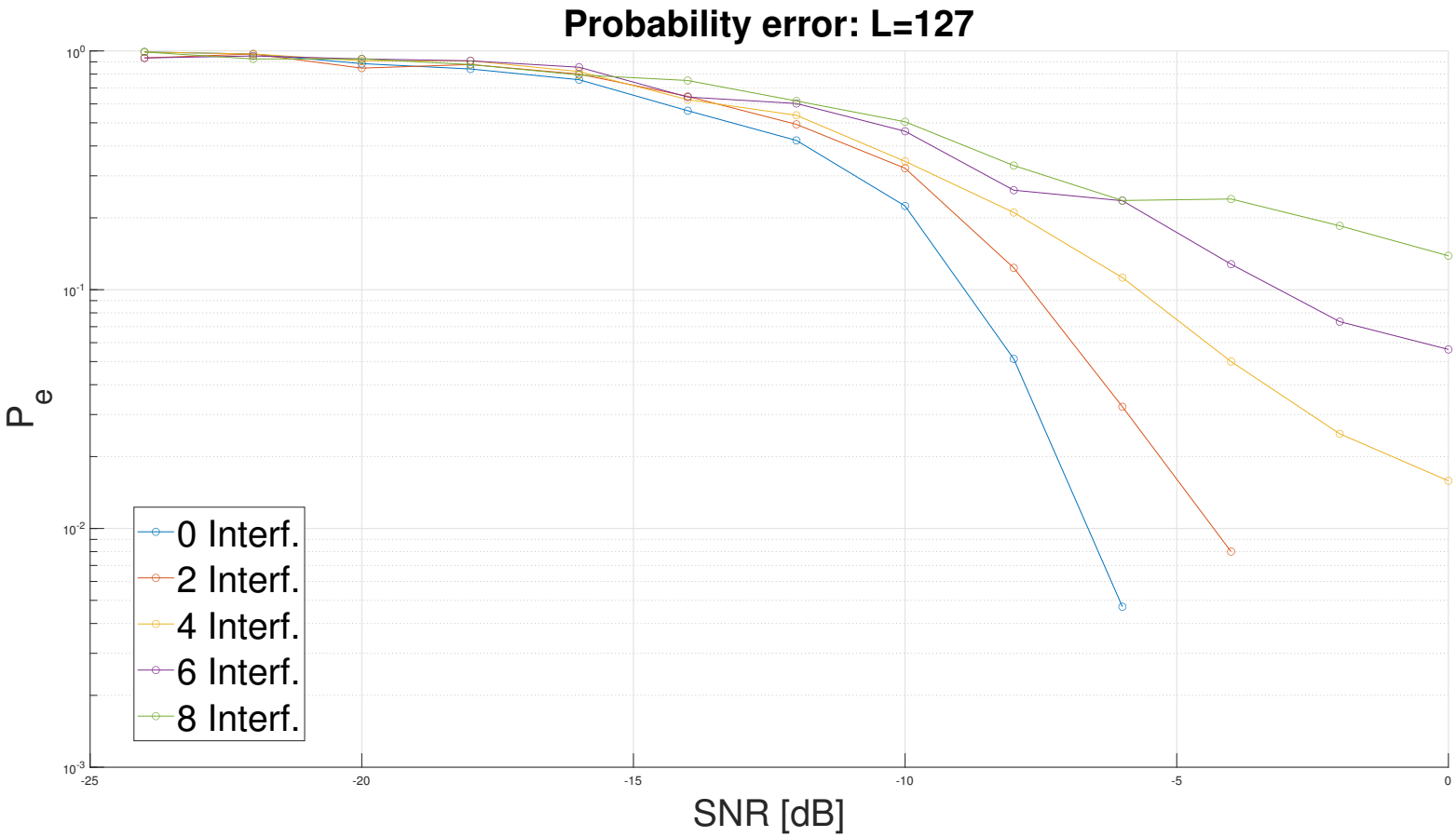
Task 2: Techniques Comparison

All three methods ultimately identify the same peak, but they differ significantly in implementation. The direct formula (.$*$ in a loop) is highly inefficient due to its $O(N^2)$ computational cost. MATLAB's `xcorr` computes linear correlation, meaning we must carefully discard the first $N-1$ points to align the positive lags correctly. Conversely, the FFT approach calculates the circular correlation directly, offering superior efficiency ($O(N \log N)$) and making it the standard choice for SDR GNSS receivers.

Task 3: Multiple Access Interference

Present a single figure with 0 2 4 6 8 interfering Gold codes.
Comment the results

Task 3: MAI with L=127



Task 3: MAI with L=127

The Vulnerability of Short Codes ($L = 127$)

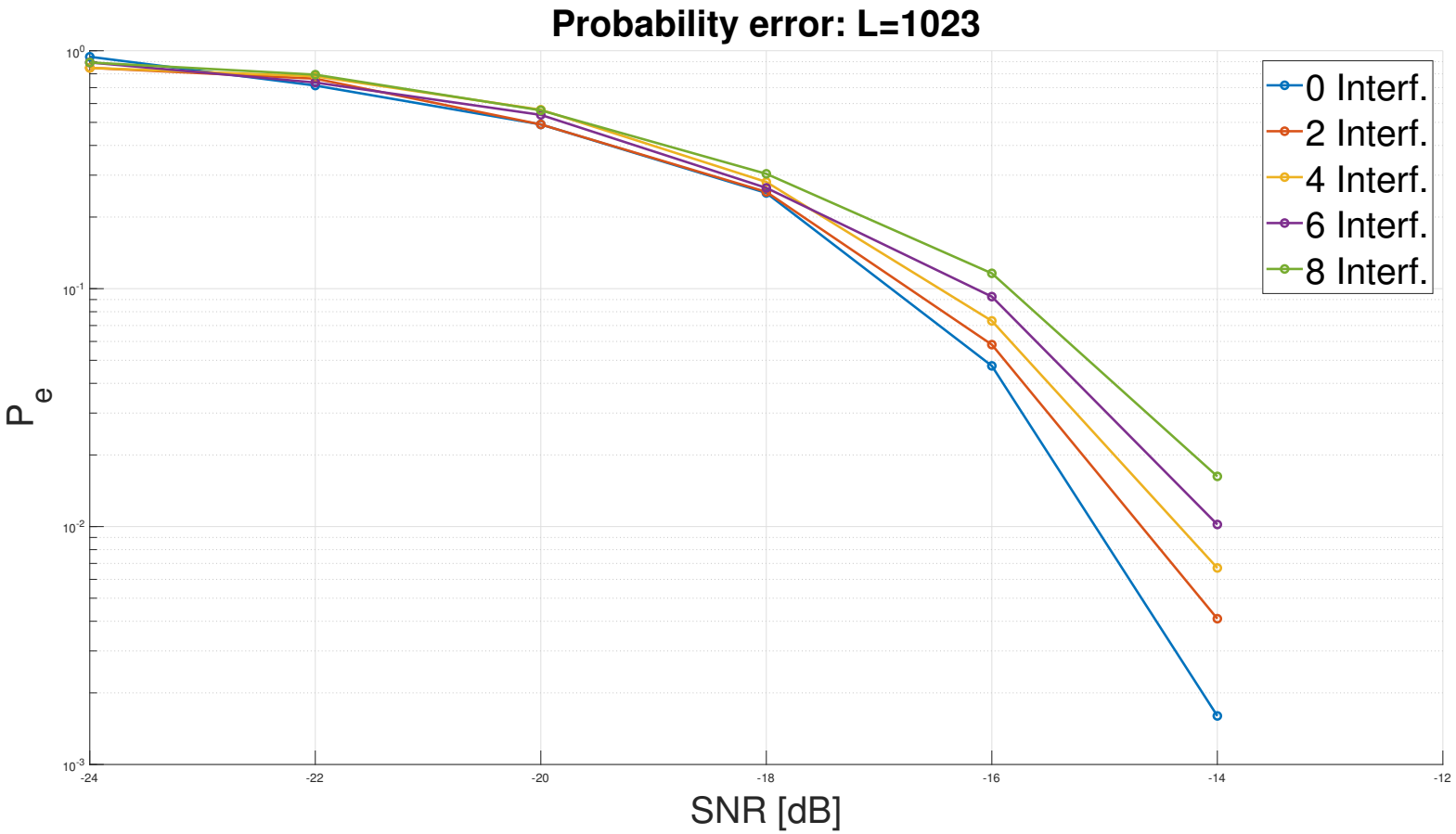
Analyzing the Probability of Error (P_e) versus SNR reveals the critical impact of sequence length on system robustness. In the $L = 127$ scenario, the receiver is highly susceptible to Multiple Access Interference (MAI). While the baseline without interference easily reaches the target error rate, introducing multiple interfering Gold codes causes the curves to fan out significantly.

Notably, with 8 interferers, the cross-correlation noise floor becomes overwhelming, preventing the system from even reaching a 10^{-1} error rate regardless of how much the SNR increases.

Task 3: Sequence Length Comparison

Present the new figure. Compare and comment the results taking into account the considerations of Task 1

Task 3: Processing Gain with L=1023



Task 3: Processing Gain with L=1023

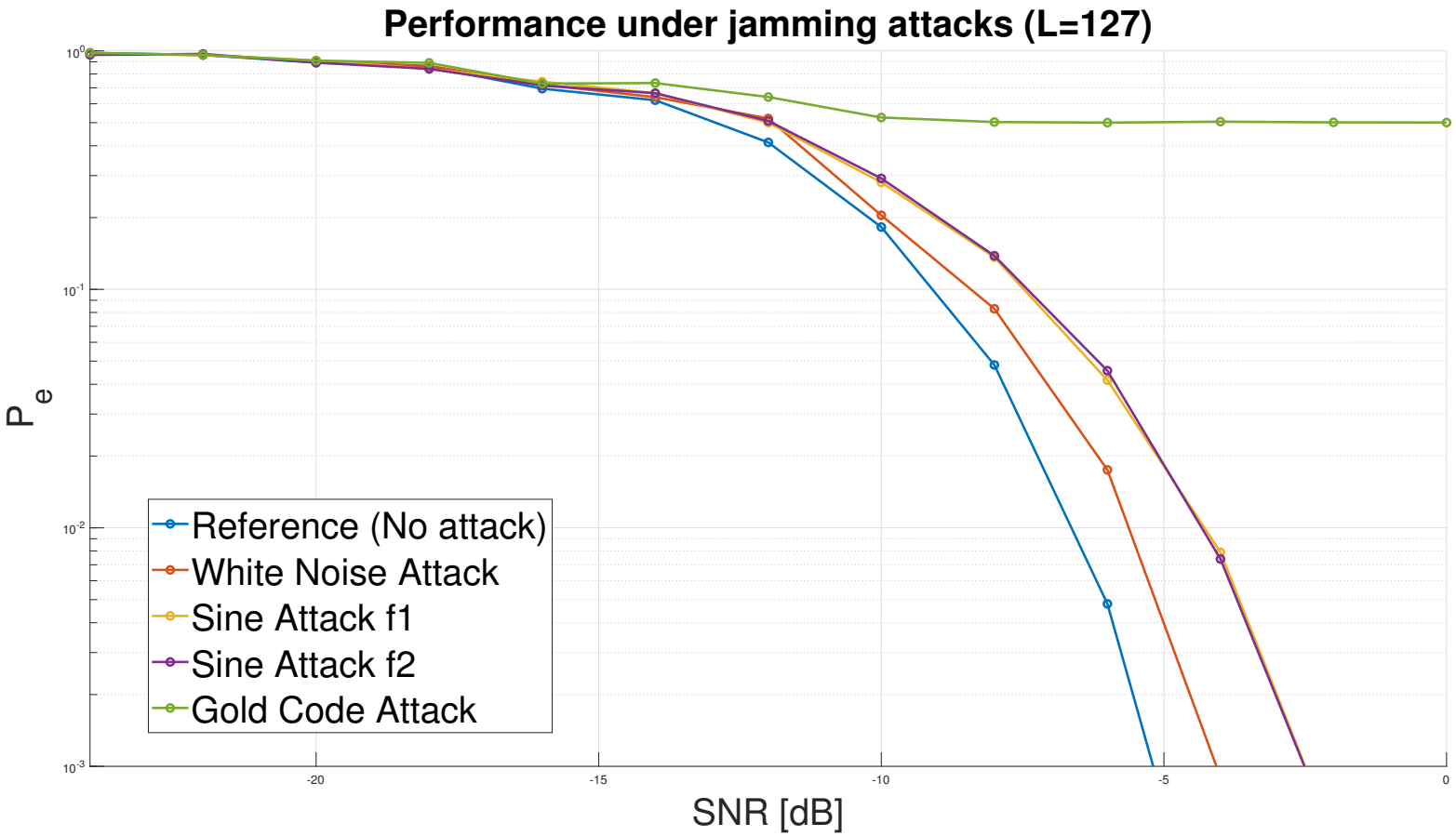
The Processing Gain Advantage ($L = 1023$)
Extending the code length to $L = 1023$ yields a drastic performance improvement. All error curves shift significantly toward lower SNR values, meaning the receiver can successfully acquire the signal even when deeply buried in noise. Furthermore, the curves remain tightly clustered; the system demonstrates remarkable resilience, suffering only a minimal performance penalty even when operating alongside 8 interfering satellites.

Validating Task 1 Theoretical Findings
These simulation results perfectly validate our theoretical observations from Task 1. The $L = 1023$ sequence inherently provides a vastly superior processing gain due to its higher ratio between the auto-correlation peak (1023) and the maximum cross-correlation bound (65). This enhanced isolation effectively suppresses the interference noise floor.

Task 4: Jamming Attacks

Show a figure with all the curves. Comment the results

Task 4: Jamming Attacks



Task 4: Jamming Attacks Analysis

- **White noise and Sine attacks:** The curves representing the white noise attack and the sinusoidal tones (f1 and f2) show a relatively marginal performance degradation compared to the "No attack" baseline. This occurs because the correlation process at the receiver effectively "spreads" the energy of the narrowband interferer across the entire spectrum, heavily mitigating its impact on peak detection.

- **Gold Code attack:** The real critical issue arises with the "Gold Code Attack". In this scenario, the error rate curve remains almost flat and extremely high regardless of the SNR. Since the attacker broadcasts the exact same Gold sequence as the target signal, the correlator is unable to distinguish between the legitimate and malicious signals, ultimately leading to tracking failure.

Task 5: Doppler Impact Overview

What is Doppler?

The Doppler effect is the apparent change in the frequency of a signal perceived by a receiver. If the satellite and the receiver move closer, the perceived frequency increases; if they move apart, it decreases.

Is it important for GPS?

It is an absolutely crucial factor, given that satellites travel at extremely high orbital velocities and the receiver itself might be in motion. This frequency shift must be properly estimated and removed.

Task 5: Consequences & Ambiguity Resolution

Which are the consequences for the performance?

If we fail to compensate for the Doppler effect, the auto-correlation peak drops drastically until it is completely buried in noise. Without a distinct peak, the receiver cannot lock onto the satellite and is unable to calculate the time of flight (t_i) required for 3D positioning.

What is the Cross Ambiguity Function?

This is the mathematical function we rely on to resolve the acquisition ambiguity. Instead of performing a standard 1D search solely over time, the CAF evaluates the correlation across a 2D grid encompassing both the time delay (τ) and the Doppler frequency shift (f_d). The global maximum on this map simultaneously yields our delay and Doppler estimates.